

Proposed Test Procedure

Smart Flood Detection and Monitoring System

1. Introduction

This document describes the proposed testing procedure for the Smart Flood Detection and Monitoring System. The purpose of testing is to verify that all hardware and software components function correctly both individually and as an integrated system. Testing will be conducted in three stages: component testing, integration testing, and system testing.

2. Component Testing

Each component will be tested independently before integration.

2.1 Arduino Microcontroller Test

Objective

Verify that the Arduino Uno correctly executes programs and communicates with connected devices.

Procedure

1. Upload a simple LED blink program.
2. Open the Serial Monitor and print test messages.
3. Verify stable operation for several minutes.

Expected Result

- LED blinks correctly.
 - Serial messages are displayed without errors.
 - Arduino operates continuously without resetting.
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2.2 Water Level Sensor Test

Objective

Verify that the water level sensor detects different water levels accurately.

Procedure

1. Connect the water level sensor to the Arduino.
2. Place the sensor in containers with varying water levels.
3. Record sensor readings from the Serial Monitor.
4. Compare readings at low, medium, and high water levels.

Expected Result

- Sensor readings increase as water level rises.
 - Readings change consistently and respond immediately to level changes.
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2.3 Ultrasonic Sensor Test

Objective

Verify accurate distance measurement.

Procedure

1. Place an object at known distances from the sensor.
2. Measure distances at 5 cm, 10 cm, 15 cm, and 20 cm.
3. Compare measured values with actual distances.

Expected Result

- Measured distances closely match actual distances.
 - Sensor readings remain stable and repeatable.
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2.4 Temperature Sensor Test

Objective

Verify that the temperature sensor provides accurate readings.

Procedure

1. Connect the temperature sensor to the Arduino.
2. Display temperature values on the Serial Monitor.
3. Compare readings with a standard thermometer.

Expected Result

- Temperature values are reasonable and stable.
 - Readings are close to reference measurements.
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2.5 LCD Display Test

Objective

Verify proper LCD functionality.

Procedure

1. Upload a test program that displays text.
2. Display different messages and sensor values.
3. Verify readability and screen updates.

Expected Result

- Characters are displayed correctly.
 - Screen updates occur without flickering.
 - All messages are visible and readable.
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2.6 LED Indicator Test

Objective

Verify operation of Green, Yellow, and Red LEDs.

Procedure

1. Activate each LED individually using a test program.
2. Observe brightness and operation.

Expected Result

- Green LED illuminates correctly.
 - Yellow LEDs illuminate correctly.
 - Red LED illuminates correctly.
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2.7 Power Source Selection Switch Test

Objective

Verify that the system responds correctly to the selected power source.

Procedure

1. Toggle the switch between available power settings.
2. Observe system behavior.

Expected Result

- Power source changes are detected correctly.
 - System remains operational after switching.
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3. Software Logic Testing

Objective

Verify that flood risk levels are determined correctly.

Procedure

1. Simulate Safe water level conditions.
2. Simulate Warning water level conditions.
3. Simulate Danger water level conditions.
4. Observe LCD messages and LED indicators.

Expected Result

Flood Level	LCD Display	LED Indicator
Safe	SAFE	Green LED
Warning	WARNING	Yellow LEDs
Danger	DANGER	Red LED

The displayed status and activated LED must correspond to the current water level condition.

4. Integration Testing

After all individual components pass testing, integration testing will be performed in stages.

Stage 1: Sensor Integration

Components

- Arduino Uno
- Water Level Sensor
- Ultrasonic Sensor
- Temperature Sensor

Procedure

1. Connect all sensors simultaneously.
2. Display readings in the Serial Monitor.
3. Verify that all sensors operate without conflicts.

Expected Result

- All sensors provide valid readings at the same time.
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Stage 2: Sensor and LCD Integration

Components

- Sensors
- Arduino Uno
- LCD Display

Procedure

1. Display sensor readings on the LCD.
2. Verify automatic updates of displayed information.

Expected Result

- LCD correctly displays water level, temperature, and status information.
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Stage 3: Sensor and LED Integration

Components

- Sensors
- Arduino Uno
- LEDs

Procedure

1. Simulate Safe, Warning, and Danger conditions.
2. Verify LED activation.

Expected Result

- Correct LED indicators activate according to flood level.
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Stage 4: Full System Integration

Components

- Arduino Uno
- Water Level Sensor
- Ultrasonic Sensor
- Temperature Sensor
- LCD Display
- LEDs
- Power Source Selection Switch

Procedure

1. Assemble the complete system.
2. Simulate multiple flood conditions.
3. Observe sensor readings, LCD updates, and LED responses.

Expected Result

- All components operate together correctly.
 - Flood conditions are identified accurately.
 - Display and indicators update in real time.
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5. System Testing

Continuous Operation Test

Procedure

1. Operate the system continuously for several hours.
2. Monitor sensor readings and output devices.

Expected Result

- No system crashes or unexpected resets.
 - Readings remain stable and reliable.
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Response Time Test

Procedure

1. Simulate a sudden increase in water level.
2. Measure the time required for the LCD and LEDs to respond.

Expected Result

- Status updates occur within one second.
 - Warning indicators activate within two seconds.
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6. Order of Integration Testing

The integration testing process will be performed in the following order:

1. Arduino Microcontroller Testing
 2. Water Level Sensor Testing
 3. Ultrasonic Sensor Testing
 4. Temperature Sensor Testing
 5. LCD Display Testing
 6. LED Indicator Testing
 7. Power Source Selection Switch Testing
 8. Sensor Integration
 9. Sensor + LCD Integration
 10. Sensor + LED Integration
 11. Full System Integration
 12. System Testing and Validation
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7. Acceptance Criteria

The Smart Flood Detection and Monitoring System will be considered successful if:

- All components pass individual testing.
- Sensor readings are accurate and consistent.
- Flood levels are correctly identified.
- LCD information is displayed correctly.
- LEDs indicate the appropriate flood risk level.
- Response times meet project requirements.
- The system operates reliably during continuous testing.

